

Qingzheng Yu

Department of Physics and Astronomy
University of Florence
Firenze, Italy, 51024

Email: qingzheng.yu@unifi.it
Website: <https://qz-yu.github.io>
ORCID: 0000-0003-3230-3981

EDUCATION

Ph.D. in Astrophysics

Sep. 2017 - Jun. 2023

Department of Astronomy, Xiamen University, Fujian, China

– Thesis: *On the Cold Gas Evolution of Interacting Galaxies*

– Advisor: Professor Taotao Fang

Bachelor of Science in Physics

Sep. 2012 - Jun. 2016

Department of Physics, Guangzhou University, Guangdong, China

PROFESSIONAL APPOINTMENTS

Postdoctoral Research Fellow

Sep. 2024 - present

Department of Physics and Astronomy, University of Florence, Italy

Postdoctoral Research Assistant

Jul. 2023 - Sep. 2024

Department of Astronomy, Xiamen University, Fujian, China

Research Assistant

Sep. 2017 - Jun. 2023

Department of Astronomy, Xiamen University, Fujian, China

RESEARCH INTERESTS

Galaxy Formation and Evolution • Circumgalactic Medium • Interstellar Medium,
Gas Recycling and Star Formation of Galaxies • Galaxy Interactions and Mergers,
Feeding and Feedback of AGN • HI Absorption Survey toward Radio AGNs

ACCEPTED PROPOSALS

PI, ~182 h, Five-hundred-meter Aperture Spherical radio Telescope (FAST)

2020-2025

PID: PT2020_0152, PT2021_0067, PT2022_0090, PT2023_0004, PT2023_0021, PT2024_0216, PT2024_0244

[Yu et al., 2023, ApJ, 952, 144](#)

[Yu et al., 2022, ApJ, 934, 114](#)

PI, 19.7 h, Karl G. Jansky Very Large Array (JVLA)

2023A

Projects: 23A-236

PI, 57 h, Institut de Radioastronomie Milimetrique (IRAM) 30 m

2021-2022A

Projects: E01-21, 029-22 [Yu et al., 2024, ApJS, 273, 2](#)

PI, 21 h, James Clerk Maxwell Telescope (JCMT)

2021B

Projects: M21BP051 [Yu et al., 2024, ApJS, 273, 2](#)

PI, 4 nights, P200/Hale Telescope: CWI, WaSP

TAP 2021A-2024A

Projects: CTAP2021-A0034, CTAP2021-B0019, CTAP2024-A0020

PI, 11 h, Green Bank Telescope (GBT) Projects: GBT-21A-245	2021A
Co-I, 41.4 h, Karl G. Jansky Very Large Array (JVLA) Projects: 24A-407	2024A
Co-I, ~134 h, Five-hundred-meter Aperture Spherical radio Telescope (FAST) 2019a-005-S, PT2021_0147, PT2021_0186, PT2021_0120, PT2021_0139, PT2021_0040, PT2022_0181, PT2024_0044, PT2024_0106	2019-2022

OBSERVING EXPERIENCE

Radio single-dish spectroscopy and mapping:

[FAST](#), H I spectral line observations of nearby galaxies, pool, 94 h
[FAST](#), H I mapping of nearby galaxy groups, pool, 34 h
[FAST](#), H I mapping of high-velocity clouds, pool, 64 h
[GBT](#), H I spectral line observations of nearby galaxies, pool, 11h
[IRAM 30m](#), CO spectral line observations of nearby galaxies, remote, 149 h
[JCMT](#), CO spectral line observations of nearby galaxies, pool, 21 h
[Parkes](#), H I spectral line observations of nearby galaxies, remote, 22 h

Radio interferometer:

[JVLA](#), H I spectral line and continuum imaging of nearby galaxies, pool, 19.7 h

Optical imaging and spectroscopy:

[P200/Hale](#), narrow-band imaging of nearby galaxies with WaSP, remote, 2 nights
[P200/Hale](#), spectroscopic observations of nearby galaxies with DBSP, remote, 1 night

SELECTED CONFERENCE CONTRIBUTION

H I Absorption in Low-power Radio AGNs Detected by FAST -Contributed talk on “Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback”, Hangzhou, China	10/2023
H I Observations of Low-power Radio AGNs and the Magellanic Stream with FAST -Contributed talk on “The First FAST Science Symposium”, FAST site, Guizhou, China	09/2023
H I Absorption in Low-power Radio AGNs Detected by FAST -Contributed talk on “Symposium on Multiwavelength Studies of Quasars and AGNs”, Lijiang, China	07/2023
Investigating Galaxy Interactions and AGN through H I Observations -Contributed talk on “Sixth Annual SKA Science Symposium”, Shanghai, China	06/2023
On the H I Content of MaNGA Major Merger Pairs, 2023, IAU Symposium, 373, 186 -Contributed remote talk on “IAUGA 2022 Symposium 373”, Busan, Korea	08/2022
On the H I Content of MaNGA Major Merger Pairs -Contributed talk on “Jing-Guang-Xia Astrophysics Symposium”, Xiamen, China	12/2021
H I observations of MaNGA merging galaxies and HVCs with FAST -Contributed talk on “CRAFTS and FAST data analysis workshop”, Nanjing, China	04/2021

TEACHING AND MENTORING

Teaching at Xiamen University

Observational Astronomy (Graduate course), Lecturer (remote) , ~30 students	2025 Spring
Observational Astronomy (Graduate course), Lecturer (1 week) , ~30 students	2024 Spring
Observational Astronomy (Graduate course), Lecturer (1 week) , ~30 students	2023 Spring
• Introduced the H I and extragalactic observations in radio band	
Observational Astrophysics (undergraduate course), Teaching Assistant, ~50 students	2018 Spring
• Built course websites for problem sets, solutions, and FAQ's for sessions	

Student Research Project Supervising

Shuo Zhang (XMU '24): Observations of Complex A with FAST, co-advisor	2022 - 2024
Zeqian Xu (XMU '25): H I Observations with FAST, co-advisor	2022 - present
Yang Su (XMU '26): A FAST Survey of H I Absorption Lines, co-advisor	2023 - present
Yuxin Xiao (XMU '25): HVC Observations with FAST, co-advisor	2022 - present

COMMUNITY SERVICE

Review panel, FAST medium-size proposal,	2025
Committee Member, FAST User Committee, ~20 out of ~500 users	2021 - present
Organizer, student discussion with colloquium speaker, Xiamen University	2021 - 2024
Organizer, group meeting for the Galaxy and Cosmology Group, Xiamen University	2020 - 2024

SKILLS

Languages:	Chinese (Mother Language), English (Professional), Italian (Beginner)
Programming:	Python, IDL, Bourne Shell, Matlab
Software & Tools:	GILDAS, HIFAST, Starlink, MIRIAD, IRAF, CWITools, GBTIDL, CASA

AWARDS/FELLOWSHIPS

2024-2026:	European Research Council (ERC) postdoctoral fellowship @ University of Florence
2024-2027:	Kavli Institute of Astronomy and Astrophysics (KIAA) fellow (declined)
2013:	Undergraduate scholarship from NAOC

REFERENCES

- Taotao Fang
Email: fangt@xmu.edu.cn
Professor of Astronomy & Astrophysics, Department of Astronomy, Xiamen University
- Cong Kevin Xu
Email: coxu@ipac.caltech.edu
Professor of Astronomy & Astrophysics,
Chinese Academy of Sciences South America Center for Astronomy,
National Astronomical Observatories, Chinese Academy of Sciences

- Enrico Di Teodoro

Email: enrico.diteodoro@unifi.it

Professor of Astronomy & Astrophysics, Department of Physics and Astronomy, University of Florence

- Junfeng Wang

Email: jfwang@xmu.edu.cn

Professor of Astronomy & Astrophysics, Department of Astronomy, Xiamen University

PUBLICATIONS

- 1st-/corresponding-authored:

6. Su Y., **Yu Q.**[†], Fang T., Wang J., Wu J., Zhang B., *A FAST Survey of HI Absorption in Low-power Radio Sources*, [2026, ApJ, 1000, 108](#)
5. Yan S., Ma A., **Yu Q.**[†], Fang T., He C., Zhu M., *HI Content of Group Galaxies from the FAST All Sky HI Survey*, [2026, ApJ, 997, 254](#)
4. **Yu Q.**, Xu C.~K., Feng S., Feng S., Gao Y., Jiang X.-J., et al., *CO Observations of Early-mid Stage Major-mergers in MaNGA Survey*, [2024, ApJS, 273, 2](#)
3. **Yu Q.**, Fang T., Wang J., Wu J., *HI Absorption in Low-power Radio AGNs Detected by FAST*, [2023, ApJ, 952, 144](#)
2. **Yu Q.**, Fang T., Feng S., Zhang B., Xu C.~K., Wang Y., Hao L., *On the HI Content of MaNGA Major Merger Pairs*, [2022, ApJ, 934, 114](#)
1. **Yu Q.** et al., *Unveiling the Interaction between the Magellanic Stream and the Milky Way's Circumgalactic Medium*, in preparation.

- Co-authored:

8. Di Teodoro E. M., Heyer M., Krumholz M. R., Armillotta L., +7+ **Yu Q.**, [2026, A&A, 706, A330](#)
7. Zhang Z., Zhang X., Li H., Fang T., +5+ **Yu Q.**, et al., [2025, ApJ, 991, 170](#)
6. Hao L., Li Z., Shen F., Xu Y., Huang Y., Lee K., **Yu Q.**, et al., [2025, ApJ, 983, 63](#)
5. Cao T.-W., Li Z.-J., Chen P.-B., Zhang C.-Y., Galaz G., Cheng C., **Yu Q.**, et al., [2024, ApJ, 971, 181](#)
4. He C., Xu, C., Lisenfeld, U., Dai, Y.~S., Fang T., Huang J., Wang W., **Yu Q.**, [2024, RAA, 24, 055005](#)
3. Zhang Z., Zhang X., Li H., Fang T., **Yu Q.**, et al., *Low- and High-velocity O VI in Milky Way-like Galaxies: the Role of Stellar Feedback*, [2024, ApJ, 962, 15](#)
2. Yi T., +17+ **Yu Q.** +10, *A Dynamically discovered and characterized non-accreting neutron star-M dwarf binary candidate*, [2022, Nature Astronomy, 6, 1203](#)
1. Zhang B., Zhu M., Wu Z.-Z., **Yu Q.-Z.**, et al., *Extragalactic HI 21-cm absorption line observations with the Five-hundred-meter Aperture Spherical radio Telescope*, [2021, MNRAS, 503, 5385](#)